MidTerm Review

Machine Learning

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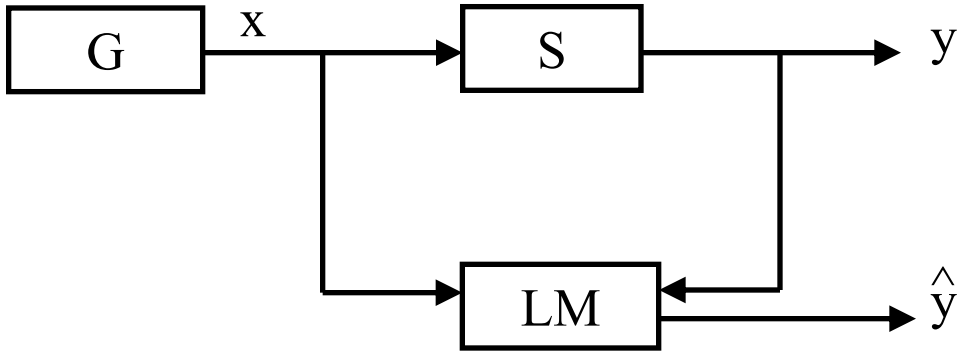
Function Estimation Model



- **Key concepts**: $F(x,y)$, an i.i.d. k-sample on F, functions $f(x,\alpha)$ and the equivalent representation of each f using its index α

The Problem of Risk Minimization

- The **loss functional** (*L, Q)*
  - the error of a given function on a given example

$$
\begin{aligned}
L:(x,y,f_\alpha) &\mapsto L(y,f(x,\alpha)) \\
Q:(z,\alpha) &\mapsto L(z_y,f(z_x,\alpha))
\end{aligned}
$$

- The **risk functional** (*R*)
  - the expected loss of a given function on an example drawn from $F(x,y)$
  - the (usual concept of) generalisation error of a given function

$$R(\alpha)=\int Q(z,\alpha)dF(z)$$

The Problem of Risk Minimization

- Three Main Learning Problems
  - Pattern Recognition:

$$y \in \{0,1\} \text{ and } L(y, f(x,\alpha)) = \mathbf{1}[y = f(x,\alpha)]$$

  - Regression Estimation:

$$y \in \Re \text{ and } L(y, f(x,\alpha)) = (y - f(x,\alpha))^2$$

  - Density Estimation:

$$y \in [0,1] \text{ and } L(p(x,\alpha)) = -\log p(x,\alpha)$$

General Formulation

- The Empirical Risk Minimization (ERM) Inductive Principle
  - Define the **empirical risk** (sample/training error):

$$R_{\text{emp}}(\alpha) = \frac{1}{k}\sum_{i=1}^{k} Q(z_i, \alpha)$$

  - Define the empirical risk minimiser:

$$\alpha_k = \arg\min_{\alpha \in \Lambda} R_{\text{emp}}(\alpha)$$

  - ERM approximates $Q(z, \alpha^*)$ with $Q(z, \alpha_k)$ the R_{emp} minimiser…that is ERM approximates α^* with α_k
  - Least-squares and Maximum-likelihood are realisations of ERM

Multiple linear regression

- Statistical Model $y = \beta_0 + \mathbf{x}^T \beta + \varepsilon$
- Model is $y_i = \beta_0 + \mathbf{x}_i^T \beta + \varepsilon_i, \quad i = 1, \cdots, \mathrm{N}$

Equivalently in matrix notation:

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$$

- $\mathbf{y}$ is N-vector of predicted values
- $\mathbf{X}$ is N $\times$ p matrix of regresses, with 1 in the first column
- β is a p-vector of parameters

Estimation by least squares

$$\hat{\beta} = \arg\min \sum_i (y_i - \beta_0 - \sum_{j=1}^{p-1} x_{ij}\beta_j)^2$$

$$= \arg\min (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

Solution is $\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

$$\hat{y} = \mathbf{X}\hat{\beta}$$

Also $Var(\hat{\beta}) = (\mathbf{X}^T\mathbf{X})^{-1}\sigma^2$

The Bias-variance tradeoff

- A good measure of the prediction performance for an estimator $\hat{f}(x)$ is the mean squared error. Let $f_0(x)$ be the true function.

$$\mathrm{MSE}[\hat{f}(x)] = \mathrm{E}[\hat{f}(x) - f_0(x)]^2$$

- This can be written as

$$\mathrm{MSE}[\hat{f}(x)] = \mathrm{Var}[\hat{f}(x)] + [E\hat{f}(x) - f_0(x)]^2$$

$$variance + \quad bias\text{^}2.$$

- When bias is low, variance is high and vice-versa.
  - Choose estimators ---- a tradeoff between bias and variance.

Linear Regression

Error Decomposition

$$
\begin{aligned}
\mathrm{Err}(x_0) &= E_T[f(x_0) - \hat{y}_0]^2 \\
&= E_T[f(x_0) - E[\hat{y}_0] + E[\hat{y}_0] - \hat{y}_0]^2 \\
&= E_T[\hat{y}_0 - E_T(\hat{y}_0)]^2 + E_T[E(\hat{y}_0) - f(x_0)]^2 \\
&\qquad + 2E\{[\hat{y}_0 - E_T(\hat{y}_0)][E(\hat{y}_0) - f(x_0)]\} \\
&= Var_T(\hat{y}_0) + Bias^2(\hat{y}_0)
\end{aligned}
$$

Regularizations

- How to reduce the variance of the models
  - Dimension reduction
  - Regularization (Ridge, LASSO)

Shrinkage methods

Ridge regression

- The ridge estimator is defined by

$$\hat{\beta}^{ridge} = \arg\min(\mathrm{Y} - \mathbf{X}\beta)^T (Y - \mathbf{X}\beta) + \lambda\beta^T\beta$$

- Equivalently,

$$\hat{\beta}^{ridge} = \arg\min(\mathrm{Y} - \mathbf{X}\beta)^T (\mathrm{Y} - \mathbf{X}\beta)$$

$$\text{subject to } \sum \beta_j^2 \le s$$

Ridge regression

- Ridge solution:

$$\hat{\beta}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathrm{Y}$$

- Singular value Decomposition:

$$\mathbf{X}=\mathbf{U}\mathbf{D}\mathbf{V}^T; \quad \mathbf{D} \text{ is a diagonal matrix with}$$

$$d_1 \geq d_2 \geq d_3 \geq \ldots \geq d_p \geq 0$$

- For Ordinary Regression

$$\mathbf{X}\hat{\beta}^{\mathrm{ls}}=\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathrm{Y}=\mathbf{U}\mathbf{U}^T\mathrm{Y}$$

Ridge regression

- Ridge solution:

$$\hat{\beta}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathrm{Y}$$

- Singular value Decomposition:

$$\mathbf{X}=\mathbf{U}\mathbf{D}\mathbf{V}^T; \quad \mathbf{D} = \mathrm{diagonal}(d_1, d_2, \ldots, d_p)$$

- For Ridge Regression

$$\mathbf{X}\hat{\beta}^{\mathrm{ridge}} = \mathbf{X}(\mathbf{X}^T\mathbf{X}+\lambda\mathrm{I})^{-1}\mathbf{X}^T\mathbf{Y}$$

$$= \mathbf{U}D\left(\mathbf{D}\mathbf{D}+\lambda\mathbf{I}\right)^{-1}D\mathbf{U}^T\mathbf{Y} = \sum_{j=1}^{p}\mathbf{u}_j\frac{d_j^2}{d_j^2+\lambda}\mathbf{u}_{\mathrm{j}}^T\mathbf{Y}$$

The Lasso

- The lasso (least absolute shrinkage and selection operator) is a shrinkage method like ridge, but acts in a nonlinear manner on the outcome y.
- The lasso is defined by

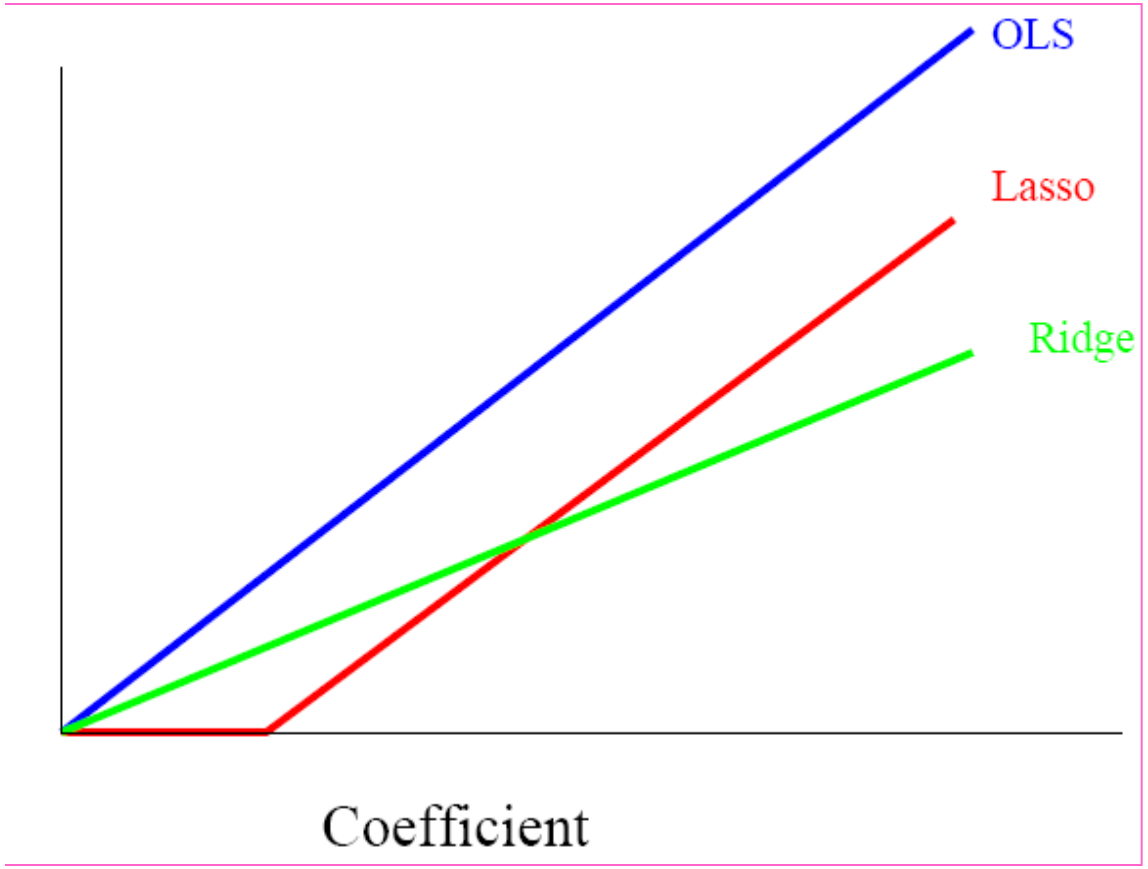
$$\hat{\beta}^{lasso} = \arg\min(\mathrm{Y} - \mathbf{X}\beta)^T(\mathrm{Y} - \mathbf{X}\beta)$$

$$\text{subject to} \quad \sum_{j=1}^{p} \left|\beta_j\right| \le t$$

- No constraint on β_0

The Lasso

- The parameter t should be adaptively chosen to minimize an estimate of expected, using say cross-validation
- *Ridge vs Lasso:* if inputs are orthogonal,
  - ridge *multiplies* least squares coefficients by a constant < 1,
  - lasso *translates* them towards zero by a constant, truncating at zero.



Question

- Prove when we increase the regularization parameter λ , the norm of $\hat{\beta}_\lambda$ will not increase.
- When we reduce the parameter t in LASSO model, the number of non-zero components in $\hat{\beta}^{lasso}$ will not increase.

Linear regression: Basis Expansion

- Base on the spline basis method: $f(x) = \sum_{k=1}^{N} \beta_k h_k(x)$

$$y = \sum_{k=1}^{m} \beta_k h_k(x) + \varepsilon \text{ , } \varepsilon \text{ is the noise.}$$

- Minimize the penalized residual sum of squares

$$RSS(f, \lambda) = \sum_{i=1}^{N} \{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 dt$$

λ is a fixed smoothing parameter

$\lambda = 0$: f can be any function that interpolates the data

$\lambda = \infty$: the simple least squares line fit

Linear regression: Basis Expansion

- The solution is a natural spline: $f(x) = \sum_{j=1}^{N} N_j(x)\theta_j$
- Then the criterion reduces to:

$$RSS(\theta, \lambda) = (y - N\theta)^T (y - N\theta) + \lambda \theta^T \Omega_N \theta$$

  – where $N = \{N_j(x_i)\ \}; \quad \Omega_{Nij} = \int N_i''(t) N_j''(t) dt$

- So the solution: $\hat{\theta} = (N^T N + \lambda \Omega_N)^{-1} N^T y$
- The fitted smoothing spline: $\hat{f}(x) = \sum_{j=1}^{N} N_j(x)\hat{\theta}_j$

Logistic Regression

- Model:

$$Log\frac{\Pr(g=1|X=x)}{\Pr(g=k|X=x)}=\beta_{1,0}+\beta_1^T x$$

$$\vdots$$

$$Log\frac{\Pr(g=K-1|X=x)}{\Pr(g=K|X=x)}=\beta_{k,0}+\beta_2^T x$$

$$\Updownarrow$$

$$\Pr(g=k|X=x)=\frac{\exp(\beta_{k,0}+\beta_k^T x)}{1+\sum_{l=1}^{K-1}\exp(\beta_{l,0}+\beta_l^T x)},\ k=1,\cdots,K-1$$

$$\Pr(g=K|X=x)=\frac{1}{1+\sum_{l=1}^{K-1}\exp(\beta_{l,0}+\beta_l^T x)}$$

Logistic Regression 2

- Parameters estimation
  - Objective function

$$\hat{\beta} = \arg\max_{\beta} \sum_{i=1}^{N} \log \Pr_{\beta}(y_i \mid x_i)$$

  - Parameters estimation

    IRLS (iteratively reweighted least squares)

    Particularly, for two-class case, using Newton-Raphson algorithm to solve the equation, the objective function:

$$p(x,\beta) = Pr_{\beta}(y=1 \mid x); \quad Pr_{\beta}(y=0 \mid x) = 1 - p(x,\beta)$$

$$p(x,\beta) = \Pr_{\beta}(y=1 \mid x) = \frac{\exp(\beta^T x)}{1+\exp(\beta^T x)};$$

$$l(\beta) = \sum_{i=1}^{N} y_i \log p(x_i,\beta) \ + (1-y_i)\log(1-p(x_i,\beta))$$

Logistic Regression 3

- Prababilty density function

$$p(x,\beta) = \Pr_\beta(y=1\,|\,x) = \frac{\exp(\beta^T x)}{1+\exp(\beta^T x)};$$

- The cost function:

$$l(\beta) = \sum_{i=1}^{N} y_i \log p(x_i,\beta) \;\; + (1-y_i)\log(1-p(x_i,\beta))$$

$$= \sum_{i=1}^{N} \{y_i \beta^T x_i - \log(1+e^{\beta^T x_i})\}$$

- The gradients

$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^{N} x_i (y_i - p(x_i;\beta)) = 0$$

$$\frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^T} = -\sum_{i=1}^{N} x_i x_i^T p(x_i;\beta)(1-p(x_i;\beta))$$

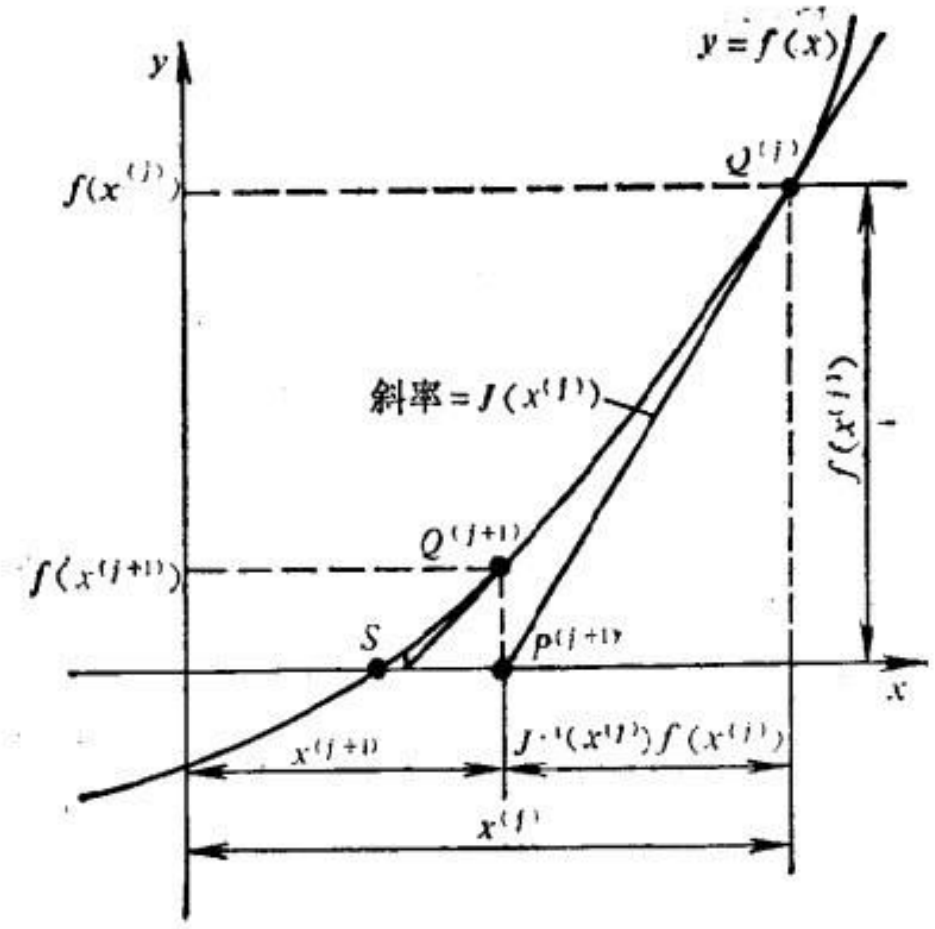
Logistic Regression 4

$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^{N} x_i (y_i - p(x_i;\beta)) = 0$$

$$\frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^T} = -\sum_{i=1}^{N} x_i x_i^T p(x_i;\beta)(1 - p(x_i;\beta))$$

The Newton Iterative method

$$\beta^{new} = \beta^{old} - \left(\frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^T} \right)^{-1} \frac{\partial l(\beta)}{\partial \beta}$$



一维牛顿-拉夫森算法的几何解释

Question

- How to estimate the parameters in logistic regression model?

Linear Discriminant Analysis

- According to the Bayes optimal classification mentioned in chapter 2, the posteriors is needed.

post probability : $\Pr(G \mid X)$

Assume:

$f_k(x)$ —— condition-density of $\mathbf{X}$ in class $G=k$.

π_k —— prior probability of class k, with $\sum_{k=1}^{K} \pi_k = 1$

Bayes theorem give us the discriminant:

$$\Pr(G = k \mid X = x) = \frac{f_k(x)\pi_k}{\sum_{l=1}^{K} f_l(x)\pi_l}$$

Linear Discriminant Analysis

- Multivariate Gaussian density:

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)}$$

- Comparing $pdfs$ of two classes k and l , assume $\Sigma_k = \Sigma, \forall k$

$$\log \frac{\Pr(G=k \mid X=x)}{\Pr(G=l \mid X=x)} = \log \frac{f_k(x)}{f_l(x)} + \log \frac{\pi_k}{\pi_l}$$

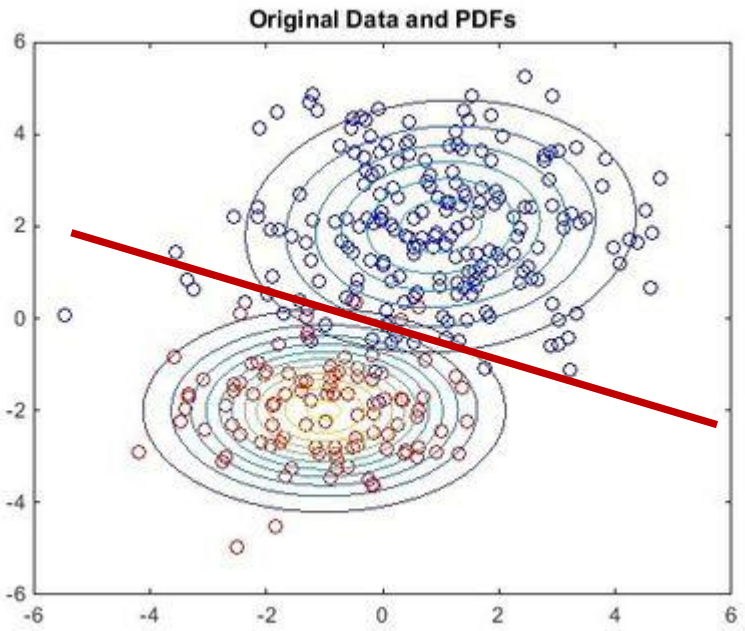
$$= \log \frac{\pi_k}{\pi_l} - \frac{1}{2}(\mu_k + \mu_l)\Sigma^{-1}(\mu_k - \mu_l)$$

$$+ x^T \Sigma^{-1}(\mu_k - \mu_l)$$

Q&A

- 关于LDA的分界面：
  - 对于线性判别分析，两类协方差相同，第一类样本数N1大约第二类样本数，分界面里哪一类中心更近？

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k \Sigma^{-1} \mu_k + \log \pi_k$$



Reduced Rank LDA, Fisher's formulation

- Find $Z = a^T X \quad s.t.$ "between-class var" is maximized relative to "within-class var".

- Maximize "Rayleigh quotient" :

$$\max\left(\frac{a^T B a}{a^T W a}\right) \qquad \text{or} \qquad \max(a^T B a), \quad s.t.\ a^T W a = 1$$

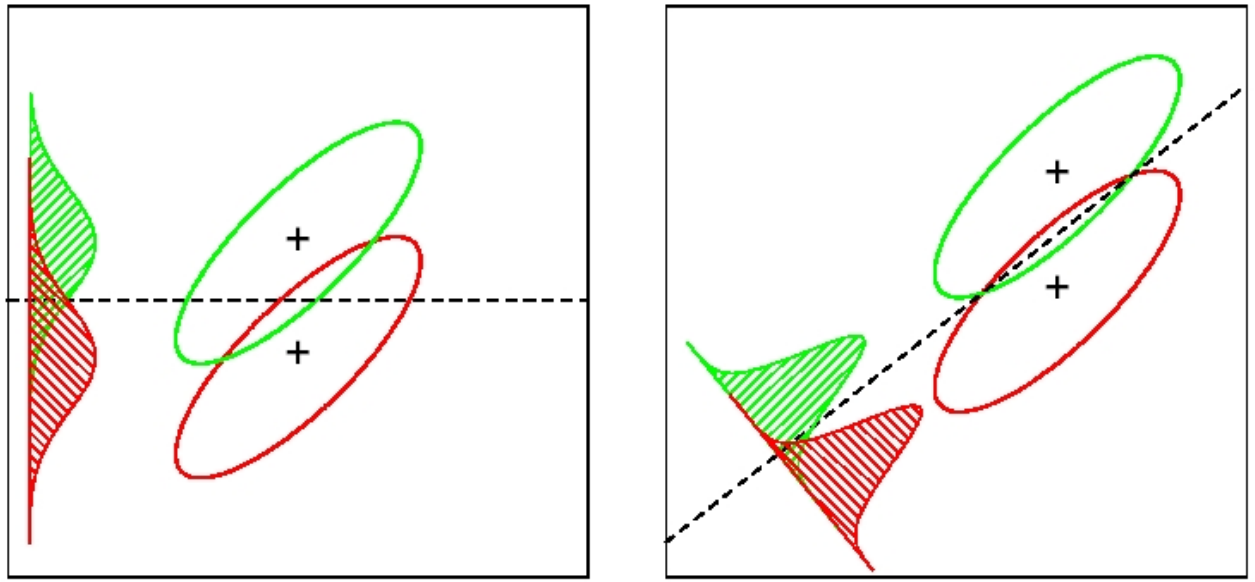
$$a = v_1 \ !$$

$$\rightarrow \max(a_2{}^T B a_2), \quad s.t.\ a_2{}^T W a_2 = 1$$

$$a_2{}^T W a_1 = 0$$

$$a = v_2 \quad \text{etc.}$$

Reduced Rank LDA: Illustration



- Although the line joining the centroids defines the direction of greatest centroid spread, the projected data overlap because of the covariance (left panel).
- The discriminant direction minimizes this overlap for Gaussian data (right panel).

Reduced Rank LDA

- Let $\hat{\Sigma} = UDU^T$

- Let $X^* = D^{-1/2}U^T X$ i.e. $\hat{\Sigma}^{-1/2} X$

  $\hat{U}_K^* = D^{-1/2}U^T \hat{U}_K$ $\hat{\Sigma}^{-1/2} U_K$

- LDA: $\delta_K(x) = \frac{1}{2}\left\| x^* - \hat{\mu}_K^* \right\|^2 - \log \hat{\pi}_K$

  - Closest centroid in sphered space(apart from $-\log \hat{\pi}_K$)

- Can project data onto K-1 dim subspace spanned by $\hat{U}_1^*, \cdots, \hat{U}_{K-1}^*$, and lose nothing!
- Can project even lower dim using principal components of $\hat{U}_k^*$, $k=1, \ldots, M(<K)$.

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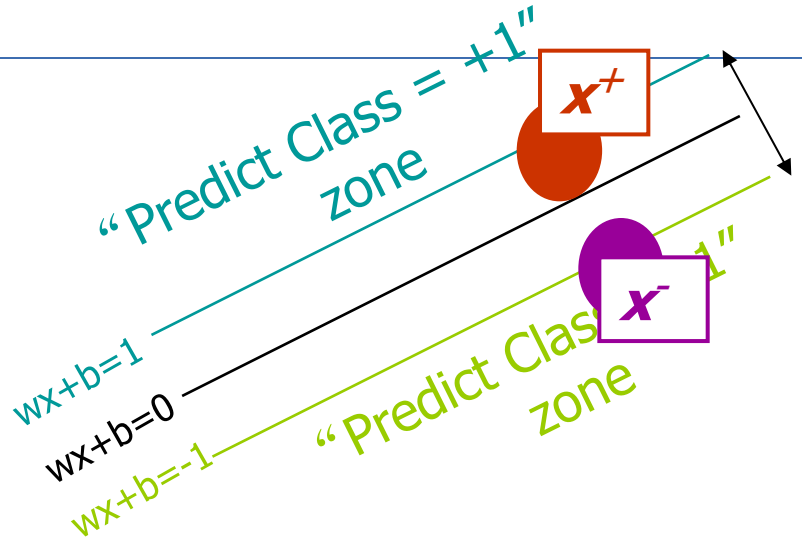
SVM: Margin Width



$$M = \text{Margin Width} = \frac{2}{\sqrt{\mathbf{w}^T\mathbf{w}}}$$

What we know:

- $\boldsymbol{w} \cdot \boldsymbol{x}^+ + b = +1$
- $\boldsymbol{w} \cdot \boldsymbol{x}^- + b = -1$
- $\boldsymbol{x}^+ = \boldsymbol{x}^- + \lambda \boldsymbol{w}$
- $|\boldsymbol{x}^+ - \boldsymbol{x}^-| = M$

$$M = |\boldsymbol{x}^+ - \boldsymbol{x}^-| = |\lambda \boldsymbol{w}| =$$

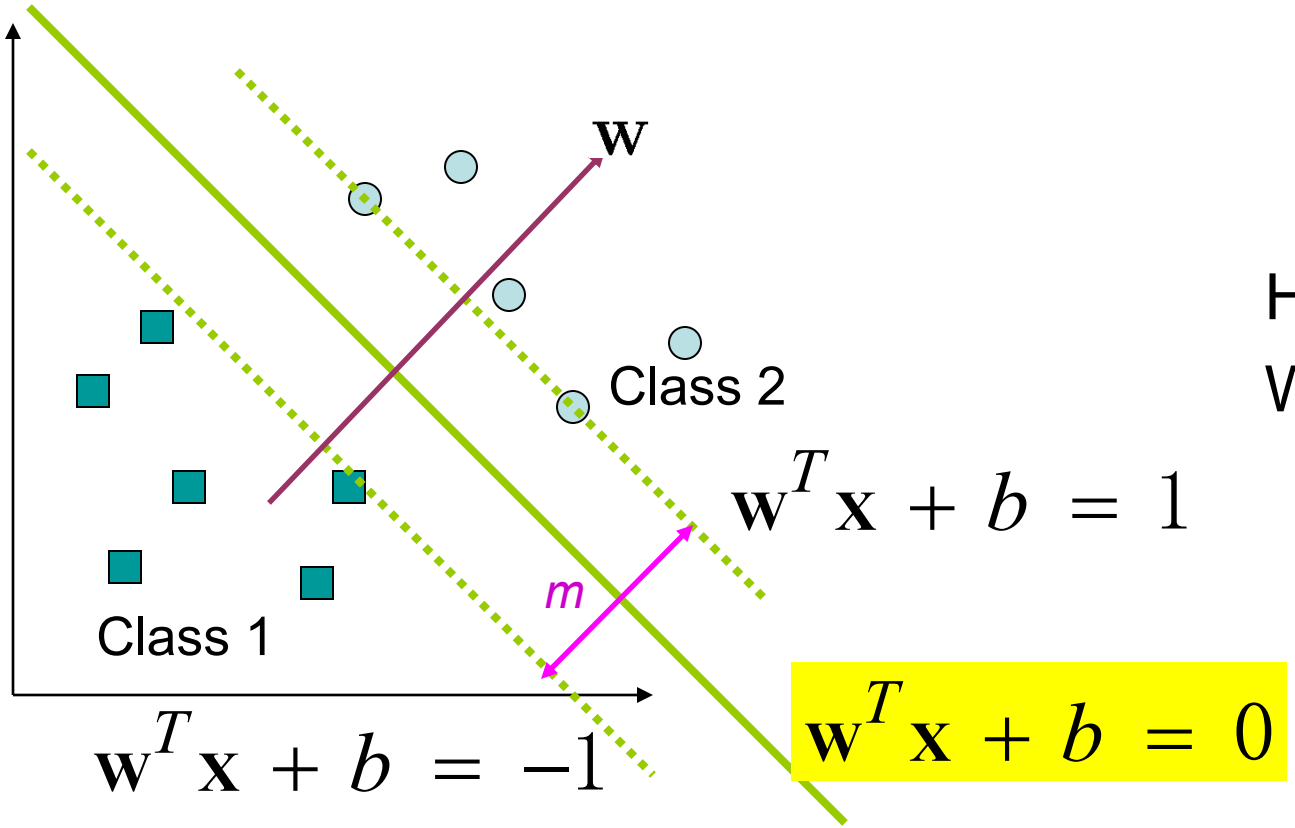
$$= \lambda |\mathbf{w}| = \lambda\sqrt{\mathbf{w}^T\mathbf{w}}$$

$$= \frac{2\sqrt{\mathbf{w}^T\mathbf{w}}}{\mathbf{w}^T\mathbf{w}} = \frac{2}{\sqrt{\mathbf{w}^T\mathbf{w}}}$$

$$\lambda = \frac{2}{\mathbf{w}^T\mathbf{w}}$$

Large-margin Decision Boundary

- The decision boundary should be as far away from the data of both classes as possible
  - We should maximize the margin, *m*
  - Distance between the origin and the line $\boldsymbol{w}^t\boldsymbol{x}=k$ is $k/\|\mathbf{w}\|$



$$m = \frac{2}{\|\mathbf{w}\|}$$

How many constraints will we have? *R*

What should they be?

$$\boldsymbol{w} \cdot \boldsymbol{x}_k + b >= 1, \quad if\ y_k = 1$$

$$\boldsymbol{w} \cdot \boldsymbol{x}_k + b <= -1, \quad if\ y_k = -1$$

*

Finding the Decision Boundary

- Let $\{x_1, ..., x_n\}$ be our data set and let $y_i \in$ {1,-1} be the class label of x_i
- The decision boundary should classify all points correctly $\Rightarrow$

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1, \qquad \forall i$$

- The decision boundary can be found by solving the following constrained optimization problem

$$\text{Minimize } \frac{1}{2}\|\mathbf{w}\|^2$$

$$\text{subject to } y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 \qquad \forall i$$

- This is a constrained optimization problem. Solving it requires some new tools
  - Feel free to ignore the following several slides; what is important is the constrained optimization problem above

Back to the Original Problem

Minimize $\|\mathbf{w}\|^2 / 2$

subject to $1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) \leq 0,$ for $i = 1, \ldots, n$

- The Lagrangian is

$$\mathcal{L} = \frac{1}{2}\mathbf{w}^T\mathbf{w} + \sum_{i=1}^{n} \alpha_i \left(1 - y_i(\mathbf{w}^T\mathbf{x}_i + b)\right)$$

  – Setting the gradient of L w.r.t. **w** and b to zero, we have

$$\mathbf{w} + \sum_{i=1}^{n} \alpha_i(-y_i)\mathbf{x}_i = \mathbf{0} \quad \Rightarrow \quad \mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

$$\sum_{i=1}^{n} \alpha_i y_i = 0$$

Illed posed problem?

*

KKT conditions

- The Karush-Kuhn-Tucker conditions,

$$\alpha_i \left[y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 \right] = 0, \quad \forall i$$

  - If $\alpha_i > 0$, then $y_i(\mathbf{w}^T \mathbf{x}_i + b) = 1$, or in other word, x_i is on the boundary of the margin;
  - If $y_i(\mathbf{w}^T \mathbf{x}_i + b) \neq 1$, x_i is not on the boundary of the margin, and $\alpha_i = 0$.

*

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The Dual Problem

- If we substitute $\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$ to $\mathcal{L}$, we have

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i^T \sum_{j=1}^{n} \alpha_j y_j \mathbf{x}_j + \sum_{i=1}^{n} \alpha_i \left(1 - y_i\left(\sum_{j=1}^{n} \alpha_j y_j \mathbf{x}_j^T \mathbf{x}_i + b\right)\right)$$

$$= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j + \sum_{i=1}^{n} \alpha_i - \sum_{i=1}^{n} \alpha_i y_i \sum_{j=1}^{n} \alpha_j y_j \mathbf{x}_j^T \mathbf{x}_i - b\sum_{i=1}^{n} \alpha_i y_i$$

$$= -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j + \sum_{i=1}^{n} \alpha_i$$

$$\sum_{i=1}^{n} \alpha_i y_i = 0$$

- This is a function of α_i only

*

The Dual Problem

- It is known as **the dual problem**: if we know $\boldsymbol{w}$, we know all α_i; if we know all α_i, we know $\boldsymbol{w}$
- The objective function of the dual problem needs to **be maximized!**

$$\text{max.} \quad W(\boldsymbol{\alpha}) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1, j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\text{subject to} \quad \alpha_i \geq 0, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0$$

Properties of α_i when we introduce the Lagrange multipliers

The result when we differentiate the original Lagrangian w.r.t. b

*

The Dual Problem

$$\text{max. } W(\boldsymbol{\alpha}) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1, j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\text{subject to } \alpha_i \geq 0, \sum_{i=1}^{n} \alpha_i y_i = 0$$

- This is a quadratic programming (QP) problem
  - A global maximum of α_i can always be found

- $\mathbf{w}$ can be recovered by

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

*

Characteristics of the Solution

- Many of the α_i are zero
  - **w** is a linear combination of a small number of data points

- $\mathbf{x}_i$ with non-zero α_i are called support vectors (SV)
  - Let t_j (j=1, ..., s) be the indices of the s support vectors. We can write

$$\mathbf{w} = \sum_{j=1}^{S} \alpha_{t_j} y_{t_j} \mathbf{x}_{t_j}$$

- For testing with a new data **z**
  - classify **z** as class 1 if the sum is positive, and class 2 otherwise

$$\mathbf{w}^T \mathbf{z} + b = \sum_{j=1}^{S} \alpha_{t_j} y_{t_j} (\mathbf{x}_{t_j}^T \mathbf{z}) + b$$

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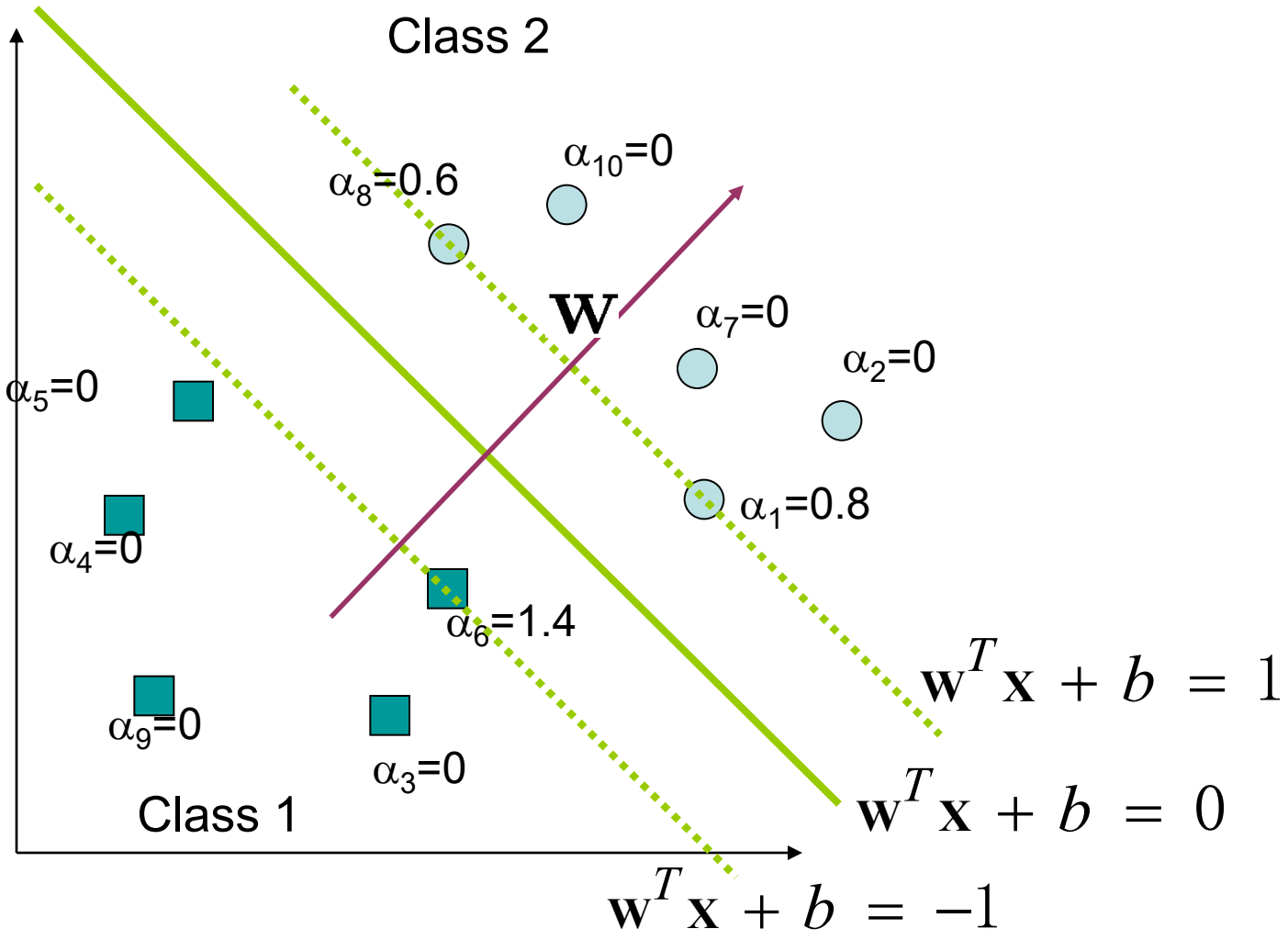
A Geometrical Interpretation



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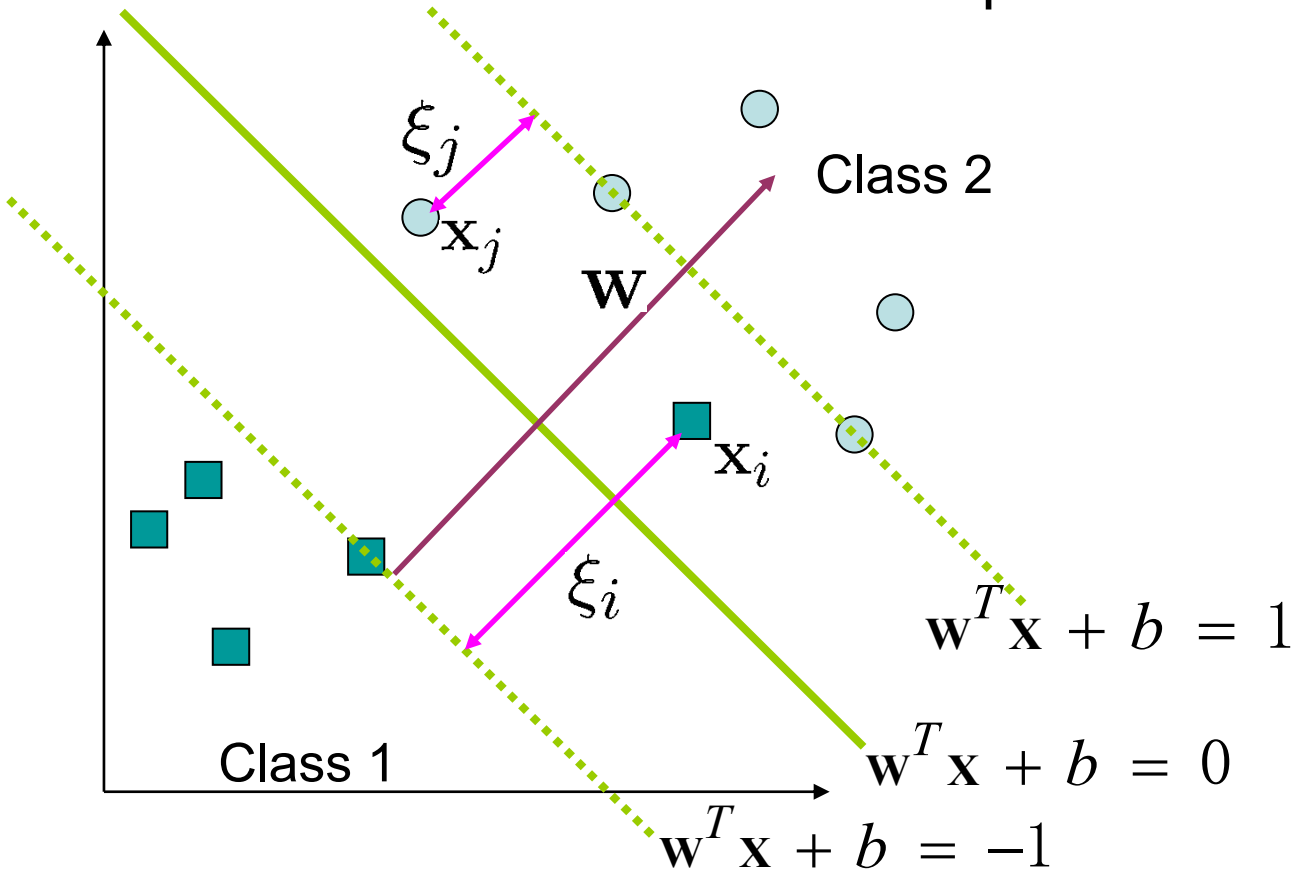
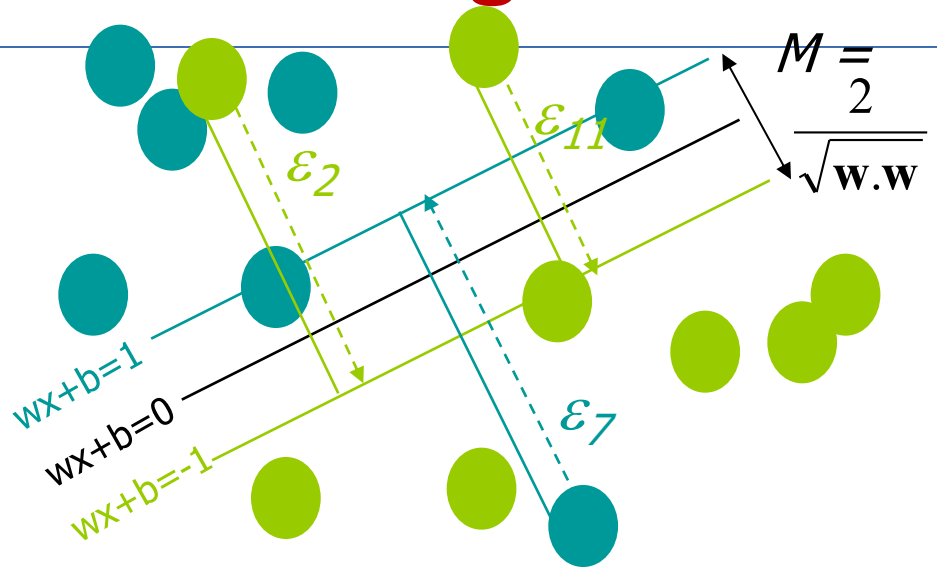
Non-linearly Separable Problems

- We allow "error" ξ_i in classification; it is based on the output of the discriminant function $\boldsymbol{w}^T\boldsymbol{x}+b$
- ξ_i approximates the number of misclassified samples



Learning Maximum Margin with Noise



The quadratic optimization criterion:

Minimize

$$\frac{1}{2}\mathbf{w}.\mathbf{w} + C\sum_{k=1}^{R}\varepsilon_k$$

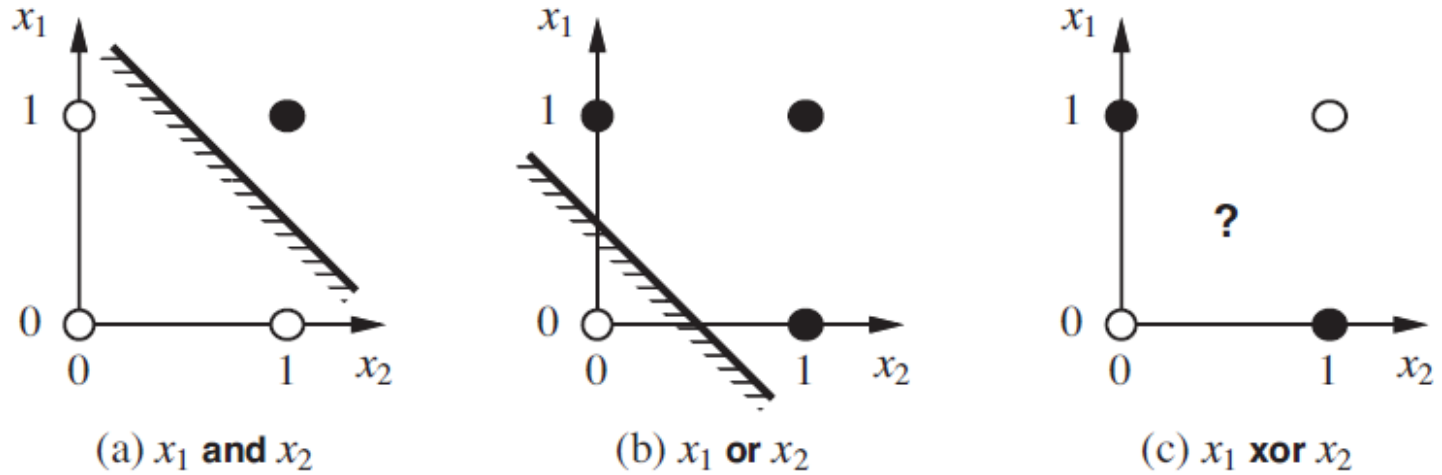
How many constraints will we have? *R*

Constraint:

$$\mathbf{w}^T\mathbf{x}_k + b \geq 1-\varepsilon_k, \quad \text{if } y_k = 1$$

$$\mathbf{w}^T\mathbf{x}_k + b \leq -1 + \varepsilon_k, \text{if } y_k = -1$$

Nonlinear Classification Problem



Linear separability in threshold perceptrons. Black dots indicate a point in the input space where the value of the function is 1, and white dots indicate a point where the value is 0. The perceptron returns 1 on the region on the non-shaded side of the line. In (c), no such line exists that correctly classifies the inputs.

Example XOR problem revisited:

Let the nonlinear mapping be :

$$\varphi(\mathbf{x}) = \left(1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1 x_2, x_2^2\right)^T ; \mathbf{x} = (x_1, x_2)^T$$

And: $$\varphi(\mathbf{x}_i) = \left(1, \sqrt{2}x_{i1}, \sqrt{2}x_{i2}, x_{i1}^2, \sqrt{2}x_{i1}x_{i2}, x_{ii2}^2\right)^T$$

Therefore the feature space is in 6D with input data in 2D

$\underline{x}_1 = (-1,-1), \quad d_1 = -1$

$\underline{x}_2 = (-1, 1), \quad d_2 = 1$

$\underline{x}_3 = (1,-1), \quad d_3 = 1$

$\underline{x}_4 = (-1,-1), \quad d_4 = -1$

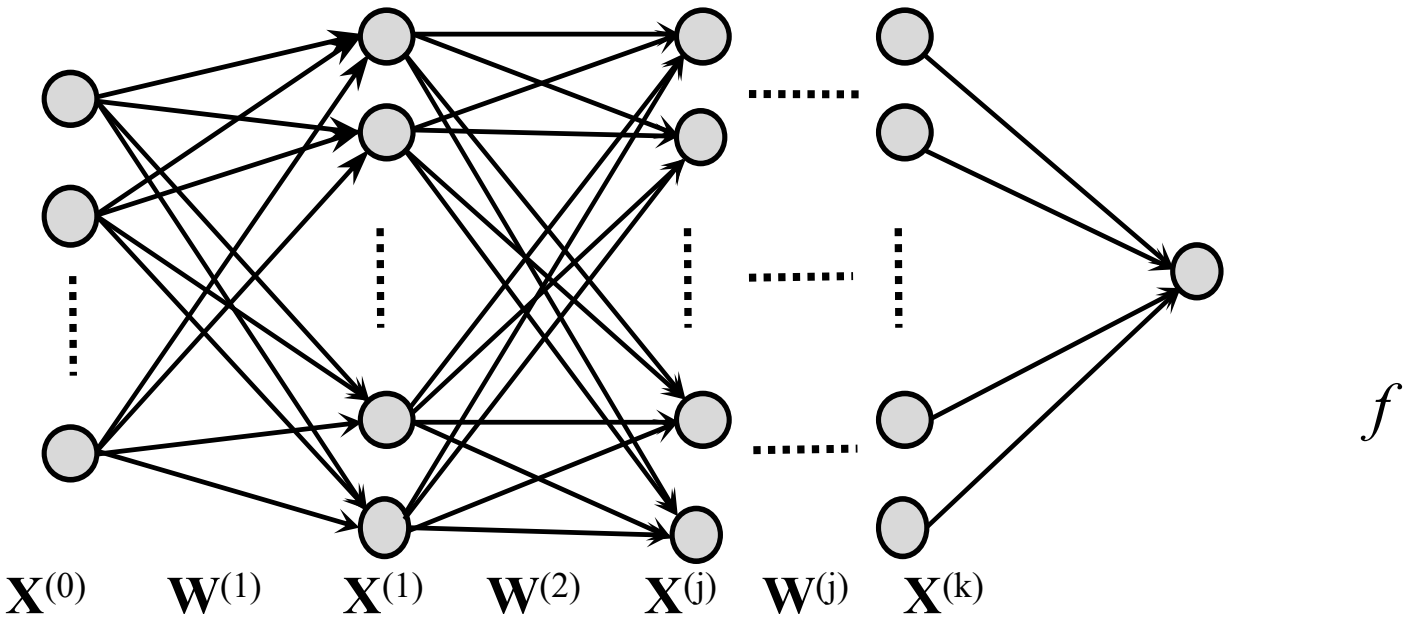
Question

- What is the advantages of kernel methods for nonlinear classification?

Multi-lawer Neural Networks

- Hidden unit $\mathbf{X}^{(j)}$: neurons in all but the last layer
- Output of *j*-th layer: $\mathbf{X}^{(j)} \rightarrow$ input of $(j+1)$-th layer
- Input vector: $\mathbf{X}^{(0)}$ Final output: f
- The weight of i-th sigmoid unit in the j-th layer: $\mathbf{W}_i^{(j)}$
- Number of sigmoid units in *j*-th layer: m_j

$$s_i^{(j)} = \mathbf{X}^{(j-1)} \cdot \mathbf{W}_i^{(j)}$$



f

The Backpropagation Algorithm

$$s_i^{(j)} = \mathbf{X}^{(j-1)} \cdot \mathbf{W}_i^{(j)}$$

- Gradient of $\mathbf{W}_i^{(j)}$:

$$\frac{\partial \varepsilon}{\partial \mathbf{W}_i^{(j)}} = \frac{\partial \varepsilon}{\partial s_i^{(j)}} \frac{\partial s_i^{(j)}}{\partial \mathbf{W}_i^{(j)}} = \frac{\partial \varepsilon}{\partial s_i^{(j)}} \mathbf{X}^{(j-1)}$$

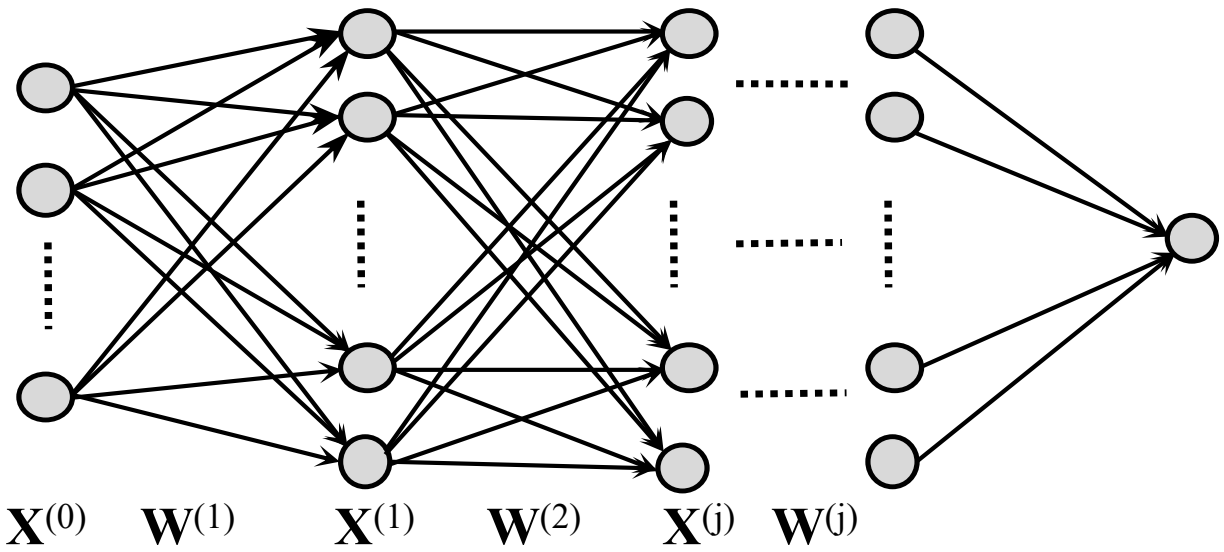
$$\frac{\partial \varepsilon}{\partial s_i^{(j)}} = \frac{\partial (d-f)^2}{\partial s_i^{(j)}} = -2(d-f)\frac{\partial f}{\partial s_i^{(j)}}$$

$$= -2(d-f)\frac{\partial f}{\partial s_i^{(j)}} \mathbf{X}^{(j-1)} = -2\delta_i^{(j)} \mathbf{X}^{(j-1)}$$

Local gradient

- Weight update:

$$\mathbf{W}_i^{(j)} \leftarrow \mathbf{W}_i^{(j)} + \eta \delta_i^{(j)} \mathbf{X}^{(j-1)}$$



Weight Update in Hidden Layers (cont.)

- Attention to **recursive equation** of local gradient!

$$\delta^{(k)} = f(1-f)(d-f)$$

$$\delta_i^{(j)} = f_i^{(j)}(1-f_i^{(j)})\sum_{l=1}^{m_j+1}\delta_i^{(j+1)}w_{i.l}^{(j+1)}$$

- Backpropagation:
  - Error is back-propagated from the output layer to the input layer
  - Local gradient of the latter layer is used in calculating local gradient of the former layer.

Questions

- What is the conditions of multi-lawer neural network which has the universal approximation?
- What is the overfitting problem in model training? What are the reasons? How to deal with the overfitting problem in shallow NN?
- What is model regularization? What are the methods of model regularization?
- Why do modern neural network models tend to adopt a deep and narrow structure rather than a shallow and wide one?
- What are the commonly used activation functions and loss functions for the nodes in the hidden layers and output layer of a neural network model?
- Why the deep neural networks are able to deal with the smal shift invariance in object recognition?

Generalization and Model Selection

- Statistical decision theory
- The Bias-Variance Decomposition
- In-Sample Prediction Error
- Model Selection Criteria
  - AIC, BIC, MDL
- Vapnik-Chernovenkis Dimension
- Cross-Validation

Regression Function

$$
\begin{aligned}
EPE(f) &= E[Y - f(X)]^2 \\
&= \int (y - f(x))^2 pr(dx, dy) \\
&= \int (y - f(x))^2 pr(dy \mid dx) pr(dx) \\
&= E_X E_{Y|X} \left([Y - f(X)]^2 \mid X \right)
\end{aligned}
$$

- 对EPE逐点极小化得： $f(x) = \arg\min_c E_{Y|X} \left([Y - c]^2 \mid X = x \right)$
- 极小解为： $f(x) = E(Y \mid X = x)$

Bias-Variance Decomposition

- Basic Model: $Y = f(X) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2)$
- The expected prediction error of a regression fit $\hat{f}(X)$

$$
\begin{aligned}
Err(x_0) &= E[(Y - \hat{f}(x_0))^2 \mid X = x_0] \\
&= \sigma_\varepsilon^2 + [E\hat{f}(x_0) - f(x_0)]^2 + E[E\hat{f}(x_0) - \hat{f}(x_0)]^2 \\
&= \sigma_\varepsilon^2 + Bias(\hat{f}(x_0))^2 + Var(\hat{f}(x_0)) \\
&= \text{Irreducible Error} + \text{Bias}^2 + \text{Variance}
\end{aligned}
$$

- The more complex the model, the lower the (squared) bias but the higher the variance.

Bias-Variance Decomposition

- For the k-NN regression fit the prediction error:

$$Err(x_0) = E[(Y - \hat{f}(x_0))^2 | X = x_0]$$
$$= \sigma_\varepsilon^2 + [f(x_0) - \frac{1}{k}\sum_{j=1}^{k} f(x_j)]^2 + \sigma_\varepsilon^2 / k$$

$$\hat{f}(x_0) = \frac{1}{k}\sum_{x_j \in N(x_0)} y_j,$$

- The in-sample error of the Linear Model

$$\frac{1}{N}\sum_{i=1}^{N} Err(x_i) = \sigma_\varepsilon^2 + \frac{1}{N}\sum_{i=1}^{N}[f(x_i) - E\hat{f}(x_i)]^2 \boxed{+ \frac{p}{N}\sigma_\varepsilon^2}$$

  – The model complexity is directly related to the number of parameters p.

Optimism of the Training Error Rate

- Training Error < True Error

$$\text{Training Error} \quad \overline{err} = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}(x_i))$$

$$\text{True Error} \quad Err = E[L(Y, \hat{f}(X)]$$

- Err is extra-sample error
- The in-sample error

$$Err_{in} = \frac{1}{N}\sum_{i=1}^{N} E_{Y^{new}}\left[L(Y_i^{new}, \hat{f}(x_i))|\mathcal{T}\right]$$

- Optimism:

$$\boxed{op \equiv Err_{in} - \overline{err}}$$

Optimism of the Training Error Rate

- The average optimism is the expectation of the optimism over training sets

$$\boxed{\omega \equiv \mathrm{E}_y(op) \equiv \mathrm{E}_y(Err_{in} - \overline{err})}$$

Training Error $\overline{err} = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}(x_i))$ In sample error: $Err_{in} = \frac{1}{N}\sum_{i=1}^{N} E_{Y^{new}}\left[L(Y_i^{new}, \hat{f}(x_i))|\mathrm{T}\right]$

$$\omega = E_y(op) = E_y\left[Err_{in} - \overline{err}\right]$$

$$= \frac{2}{N}\sum_{i=1}^{N} Cov(\hat{y}_i, y_i)$$

Model Selection: AIC & BIC

$$\mathrm{AIC} = \overline{err} + 2\frac{d}{N}\hat{\sigma}_{\varepsilon}^{2}$$

$$\mathrm{BIC} = \overline{err} + (\log N)\frac{d}{N}\sigma_{\varepsilon}^{2}$$

*

Model Selection: Cross Validation

- Denote the fitted function by $\hat{f}^{-k}(x)$ with removing k-th fold data. Then the cross-validation estimate of prediction error is

$$\mathrm{CV}(\hat{f}) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}^{-k}(x_i))$$

- The case K = N, *leave-one-out* cross-validation.
- Given model family $f(x,\alpha)$ indexed by a tuning parameter α.
- $\hat{f}^{-k}(x,\alpha)$ ： fited with the kth part of the data removed.

$$\mathrm{CV}(\hat{f},\alpha) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{f}^{-k}(x_i,\alpha))$$

Computational Learning Theory

- Maximum Likelihood
- Maximum A Posteriori，MAP
- PAC learning
  - Sample complexity
  - Computational complexity

Maximum Likelihood

- Maximizes $L(\theta)$ by solving the following equation

$$\left.\frac{\partial \ln L(\theta)}{\partial \theta}\right|_{\theta=\theta^*} = 0$$

  – Often easier to maximize $\ln L(\theta)$

  – $L(\theta)$ and $\ln L(\theta)$ achieve their maximum **at the same location**.

- $\ln L(\theta)$ converts the product into a summation.

$$\ln L(\theta) = \sum_{i=1}^{N} \ln p(x_i, \theta)$$

Log Maximum Likelihood

- The new maximization condition is:

$$\left.\frac{\partial \ln L}{\partial \theta}\right|_{\theta=\theta^*} = \sum_{i=1}^{N} \left.\frac{\partial}{\partial \theta} \ln p(x_i, \theta)\right|_{\theta=\theta^*} = 0$$

- **Resultant equations**: simple linear equations or coupled non-linear equations.

- **Example:** Let $p(x,\theta)$ be given by a Gaussian distribution function.
- Let $\theta = \mu$ be the mean of the Gaussian.
- To find the best estimate of θ from our set of n easurements .

$$\{x_1, x_2, \ \ldots, x_N\}$$

Example : Regression formulation

- Consider a linear expansion

$$y = f_\beta(x) = \sum_{j=1}^{L} \beta_j h_j(x) + \varepsilon, \qquad f(\varepsilon, \sigma^2) = \frac{1}{\sqrt{2\pi}\ \sigma} e^{-\frac{\varepsilon^2}{2\sigma^2}}$$

- Conditional probability:

$$p(y \mid x, \beta) = N(f_\beta(x), \sigma^2)$$

- The log-likelihood

$$l(\theta) = -\frac{N}{2}\log \sigma^2 2\pi - \boxed{\frac{1}{2\sigma^2}\sum_{i=1}^{N}\left(y_i - \beta^T h(x_i)\right)^2}$$

- Estimating equations: $\dfrac{\partial l(\theta)}{\partial \beta} = 0; \qquad \dfrac{\partial l(\theta)}{\partial \sigma^2} = 0.$

Maximum Likelihood and Posteriori Maximazation

- Given training data $D = \{x_i\}_{i=1}^{N}$, estimate parameter θ by maximizing its likelihood

$$\hat{\theta} = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i \mid \theta)$$

where $p(x \mid \theta)$ is its probabilty density function

- Assume that $p(\theta)$ is the prior distribution for θ . The better approach for estimating the parameter θ is Posteriori Maximazation

$$\hat{\theta} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta) p(\theta)$$

Likelihood prior

Regularization as Posteriori Maximazation

- Consider the regression problem for a parametric model

$$y = f(x \mid \theta) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2)$$

Given training data $D = \{(x_i, y_i)\}_{i=1}^N$, find the parameter θ by maximizing its likelihood

$$\hat{\theta} = \arg\max_\theta \prod_{i=1}^N p(y_i \mid x_i, \theta)$$

- If the error term follows $\varepsilon \sim N(0, \sigma_\varepsilon^2)$, then the maximum likelihood is equivalent to the RSS minimization.

- Posteriori Maximazation:

$$\hat{\theta} = \arg\max_\theta p(\theta \mid D) = \arg\max_\theta \prod_{i=1}^N p(y_i \mid x_i, \theta) p(\theta)$$

$$\boxed{= \arg\max_\theta \sum_{i=1}^N \log p(y_i \mid x_i, \theta) + \log p(\theta)}$$

An Example

- Consider a linear expansion

$$y = \mu_\beta(x) + \varepsilon = \sum_{j=1}^{N} \beta_j h_j(x) + \varepsilon; \qquad p(\varepsilon) = \frac{1}{(2\pi)^{1/2}\sigma_\varepsilon} \exp(-\tfrac{\varepsilon^2}{2\sigma^2})$$

- The prior probability

$$\beta \propto N(0, \tau \Sigma)$$

- The maximum a posterior

$$p(\beta | y, x) \approx p(\beta) p(y | x, \beta) = N(0, \tau \Sigma) N_y(\mu_\beta(x) | \sigma_\varepsilon^2)$$

MAP (Maximum A Posterior)

- The posterior distribution for β is also Gaussian, with mean and covariance

$$\mathrm{E}(\beta|\,\mathbf{Z}) = \left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\mathbf{H}^T\mathrm{y}, \qquad \mathrm{cov}(\beta|\,\mathbf{Z}) = \left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\sigma^2.$$

- The corresponding posterior values for $\mu(x) = \sum_{j=1}^{N}\beta_j h_j(x)$

$$\mathrm{E}(\mu(x)|\,\mathbf{Z}) = h(x)^T\left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\mathbf{H}^T\mathrm{Y},$$

$$\mathrm{cov}\left[\mu(x),\,\mu(x')|\,\mathbf{Z}\right] = h(x)^T\left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}h(x')\sigma^2.$$

Finite Hypothesis Case (Cosistent)

- If the hypothesis space H is finite, denote the empirical risk minimization solution as

$$\hat{h} = \arg\min_{h \in H} \hat{\varepsilon}_S(h)$$

- **Theorem:** Assume that H is a Finite Hypothesis class. For any $\varepsilon, \delta \in (0,1)$ the sample complexity is given by

$$m(\varepsilon, \delta) = \frac{\log(|H|/\delta)}{\varepsilon}.$$

- i.e. for any $m \geq m(\varepsilon, \delta)$ sample data, the empirical risk minimization solution $\hat{h} \in H$ satisfies

$$p(\varepsilon_D(\hat{h}) > \varepsilon) \leq \delta.$$

- In other words, $\hat{h} \in H$ the following inequality satisfies with at least probability $1 - \delta$

$$\varepsilon_D(\hat{h}) \leq \varepsilon.$$

Generalization Error Bound (Inconsistent)

- **Theorem.** Let $|H| = k$, and let any N, δ be fixed. Then with probability at least $1-\delta$ we have that

$$\varepsilon(\hat{h}) \le \min_{h \in H} \varepsilon(h) + 2\sqrt{\frac{1}{2N}\log\left(\frac{2k}{\delta}\right)}$$

Corollary. Let |H| = k, and let any δ, ε be fixed. Then for

$$\varepsilon(\hat{h}) \le \min_{h \in H} \varepsilon(h) + 2\varepsilon$$

to hold with probability at least $1-\delta$, it suffices that

$$N \ge \frac{1}{2\varepsilon^2}\log\left(\frac{2k}{\delta}\right) = O\left(\frac{1}{2\varepsilon^2}\log\left(\frac{k}{\delta}\right)\right)$$